# Case 2 — Mr. Yusuf Ibrahim

62 M | 10 months post-stroke — Left Cerebellar Infarct | Community Physiotherapy Clinic | Cerebellar ataxia

Target: 70%+ | Neurological Physiotherapy

## ■ PROBLEM LIST

- Cerebellar ataxia
- Dysmetria (past-pointing)
- Positive Romberg's
- Wide-based gait + trunk sway
- Near-miss falls ×2
- Strength intact

## ■ RED FLAGS

- Positive / abnormal finding: Expected score 30–44 range (fall risk zone). Specific impairments on tandem standing, eyes-closed standing, turning 360°, and single-leg stance. This baseline guides my goal: achieve BBS ≥45 as the primary measurable rehabilitation target.
- Positive / abnormal finding: Expected >15–20 seconds given ataxic gait. Observe wide-based pattern, marked hesitation and sway during the turn (highest fall risk moment). Dual-task TUG significantly slower than single-task — confirms cognitive-motor interference relevant to community environments.

## ■ PATHOPHYSIOLOGY

- 2 PATHOPHYSIOLOGY OF WORKING DIAGNOSIS What Is Happening Neurologically Mr.
- Ibrahim has sustained an ischaemic infarct of the left cerebellar hemisphere , with probable involvement of the vermis given the prominent truncal ataxia.
- The cerebellar vermis specifically coordinates axial and gait stability, and its damage produces the characteristic wide-based gait, trunk sway, and staggering steps Mr.
- Typical Signs & Symptoms Explained by Pathophysiology Sign / Symptom Pathophysiological Explanation "Boat rocking" sensation (oscillopsia) Disrupted cerebellar-vestibular integration — the cerebellum normally damps oscillatory vestibular signals.
- Without this damping, a persistent sense of motion is experienced even at rest.

## ■ DIFFERENTIAL DIAGNOSES

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## ■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

**Berg Balance Scale (BBS)**

What it measures: Static and dynamic balance across 14 functional tasks (sitting, standing, reaching, turning, tandem standing, single-leg stance). Scored 0–56; <45 indicates significantly elevated fall risk; MCID = 4 points. Why I choose it: The BBS is the gold-standard function

**Timed Up and Go Test (TUG)**

What it measures: Functional mobility — time to rise from a chair, walk 3 metres, turn, return and sit. Normative values for healthy adults aged 60+ ≈ 9–10 seconds; >13.5 seconds = significantly elevated fall risk. MCID = 2.9 seconds post-stroke. Why I choose it: The TUG specific

**Modified Clinical Test of Sensory Interaction in Balance (mCTSIB)**

What it measures: How the patient uses and integrates the three sensory systems (vision, vestibular, somatosensory/proprioception) to maintain upright balance, across four progressively challenging conditions. Why I choose it: Mr. Ibrahim's history specifically identifies worseni

## ■ TREATMENT PLAN

**■ Initial Phase — Weeks 1–4 (Community Physiotherapy Clinic)**

**■ Core Neurophysio Interventions:**

- Task-specific cerebellar coordination training (adapted Frenkel's exercises):
- Highly repetitive, progressively demanding coordination tasks targeting the specific impairments — dysmetria and trunk ataxia.
- Progressive balance retraining using sensory manipulation (mCTSIB-guided):
- Begin at the condition where the patient is just at the limit of stability; systematically progress by removing vision, challenging the surface, and adding conflicting visual backgrounds.
- Gait retraining — systematic base-of-support narrowing with directional challenge:

**■ Long-Term Plan — Weeks 4–12+ (Community Reintegration / Self-Management)**

- Graduated community re-exposure:
- Group stroke exercise class / community balance programme:
- Fully self-managed home programme by Week 8–12:
- Transition from clinic-led to independently managed programme; therapist reviews monthly then 3-monthly to reassess and progress.
- Meaningful activity and leisure reintegration:

**■ Discharge criteria:**

• Examiner Demonstration — What I Would Show in the Room
• Frenkel's heel slide / foot placement technique:
• Progressive tandem stance with sensory challenge: